To install any version of Kubernetes on an EC2 instance running Ubuntu 22.04, follow these steps:

Prerequisites

1. **EC2 Instance**: Ensure your EC2 instance is running and you have SSH access.

2. **Update Packages**: Update your system's package index.

sh

```
sudo apt-get update
```

Install Docker

Kubernetes requires a container runtime, and Docker is a common choice.

sh

```
sudo apt-get install -y apt-transport-https ca-certificates curl software-properties-common

curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo apt-key add -

sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable"

sudo apt-get update

sudo apt-get install -y docker-ce

sudo systemctl enable docker

sudo systemctl start docker
```

Install Kubernetes Components

1. **Add Kubernetes repository and key**:

sh

```
sudo apt-get install -y apt-transport-https curl

curl -s https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add -
```

echo "deb https://apt.kubernetes.io/ kubernetes-xenial main" | sudo tee /etc/apt/sources.list.d/kubernetes.list

sudo apt-get update

2. **Install kubelet, kubeadm, and kubectl**:

To install the latest version:

sh

sudo apt-get install -y kubelet kubeadm kubectl

sudo apt-mark hold kubelet kubeadm kubectl

To install a specific version, specify the version number like this:

sh

VERSION=1.24.0

sudo apt-get install -y kubelet=$VERSION-00 kubeadm=$VERSION-00 kubectl=$VERSION-00

sudo apt-mark hold kubelet kubeadm kubectl

Initialize the Kubernetes Cluster

1. **Initialize the Master Node**:

sh

sudo kubeadm init --pod-network-cidr=192.168.0.0/16

2. **Set up the kubeconfig file for the root user**:

sh

```
sudo mkdir -p $HOME/.kube

sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config

sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

Install a Pod Network Add-on

For example, to install the Flannel network add-on:

sh

```
kubectl apply -f https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml
```

Join Worker Nodes to the Cluster

To add worker nodes to your cluster, run the command that `kubeadm init` outputs at the end on each worker node. It looks something like this:

sh

```
sudo kubeadm join <master-node-ip>:6443 --token <token> --discovery-token-ca-cert-hash sha256:<hash>
```

Verify the Cluster

To check the status of your cluster, use:

sh

```
kubectl get nodes

kubectl get pods --all-namespaces
```

This will show you the nodes and pods running in your cluster.

Additional Configuration

Depending on your use case, you might want to:

1. **Set up Role-Based Access Control (RBAC)** for security.
2. **Deploy additional services** such as Ingress controllers, Metrics server, and others.

Conclusion

By following these steps, you can set up a Kubernetes cluster on an EC2 instance running Ubuntu 22.04. If you have specific versions or additional requirements, you can modify the commands accordingly.

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Running a specific version of Kubernetes in an EC2 instance involves setting up the desired version of Kubernetes components and configuring the cluster accordingly. Here's a step-by-step guide to install any specific version of Kubernetes on an EC2 instance running Ubuntu 22.04:

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2. **Update Packages**: Update your system's package index.

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```
sudo apt-get update
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sudo add-apt-repository "deb [arch=amd64] https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable"

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sudo apt-get install -y docker-ce

sudo systemctl enable docker

sudo systemctl start docker

Install Kubernetes Components

1. **Add Kubernetes repository and key**:

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curl -s https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add -

echo "deb https://apt.kubernetes.io/ kubernetes-xenial main" | sudo tee /etc/apt/sources.list.d/kubernetes.list

sudo apt-get update

2. **Install kubelet, kubeadm, and kubectl**:

To install a specific version, specify the version number like this:

sh

VERSION=1.24.0

sudo apt-get install -y kubelet=$VERSION-00 kubeadm=$VERSION-00 kubectl=$VERSION-00

sudo apt-mark hold kubelet kubeadm kubectl

Replace `1.24.0` with the desired version number.

Initialize the Kubernetes Cluster

1. **Initialize the Master Node**:

sh

sudo kubeadm init --pod-network-cidr=192.168.0.0/16

You can also specify the Kubernetes version during initialization:

sh

sudo kubeadm init --kubernetes-version $VERSION --pod-network-cidr=192.168.0.0/16

2. **Set up the kubeconfig file for the root user**:

sh

mkdir -p $HOME/.kube

sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config

sudo chown $(id -u):$(id -g) $HOME/.kube/config

Install a Pod Network Add-on

For example, to install the Flannel network add-on:

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To add worker nodes to your cluster, run the command that `kubeadm init` outputs at the end on each worker node. It looks something like this:

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To check the status of your cluster, use:

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This will show you the nodes and pods running in your cluster.

Additional Configuration

Depending on your use case, you might want to:

1. **Set up Role-Based Access Control (RBAC)** for security.
2. **Deploy additional services** such as Ingress controllers, Metrics server, and others.

Conclusion

By following these steps, you can set up and run any version of Kubernetes on an EC2 instance running Ubuntu 22.04. Adjust the version numbers and commands according to your specific requirements.